

Chetan Ghanshyam Gupta

chetan.gupta.aiml@ghrce.raisoni.net · +91-9764001278 · github.com/Chetan2025 · [linkedin.com/in/chetan-gupta-1608a8344](https://www.linkedin.com/in/chetan-gupta-1608a8344) ·
LeetCode /GFG: chetangupta2025 · <https://chetan-gupta-portfolio.vercel.app/>

EDUCATION

G.H. Rasoni College of Engineering, Nagpur

B.Tech, Computer Science & Engineering (AI & ML) | CGPA: **8.83** / 10.0

Aug 2024 – May 2028

Nagpur, India

PROJECTS

NoteLab — Academic Resource Android App

Kotlin · Android · Firebase · MVVM

Production app actively used by 100+ students at GHRCE campus

May 2026 – Present

- Designed and shipped a **production Android application** serving 100+ active users, providing organized access to lecture notes, and past-year papers with real-time sync.
- Implemented MVVM architecture with LiveData and Repository pattern, separating UI, business logic, and data layers — reducing onboarding time for new contributors by maintaining clear module boundaries.
- Integrated Firebase Realtime Database for sub-second cross-device synchronization and Firebase Auth for secure role-based login, handling concurrent reads without race conditions.
- Reduced APK cold-start time by ~28% through lazy-loading fragments, RecyclerView view recycling, and deferred Firebase initialization.

Full-Stack AI Document Chatbot Web Platform

Python · FastAPI · PostgreSQL · pgvector · JavaScript

Full-stack multi-tenant platform — custom document-aware chatbots via REST API

Apr 2025 – Jun 2025

- Built a full-stack AI platform with a FastAPI backend and JavaScript frontend enabling users to create isolated, document-aware chatbots through an **API-key-based multi-tenant architecture**.
- Implemented vector similarity search using **PostgreSQL + pgvector** with BAAI/bge-small-en-v1.5 embeddings, achieving low-latency semantic retrieval over user-specific document corpora.
- Enforced chatbot-level data isolation ensuring each API key accesses only its own indexed document corpus; deployed frontend on Vercel and backend on Render with PostgreSQL integration for one month.

Vision Transformer (ViT) Image Classifier

Python · PyTorch · Transfer Learning

End-to-end ML training pipeline — from scratch to fine-tuned model

May 2025 – Present

- Implemented ViT architecture from scratch in PyTorch — patch embedding, multi-head self-attention, positional encoding, and Reduced training time by ~80% by fine-tuning a pre-trained ViT backbone on a custom 1,200-image dataset
- Built a reproducible ML pipeline: data augmentation, learning-rate warm-up scheduling, and evaluation with per-class accuracy and confusion matrix reporting.

TECHNICAL SKILLS

- Languages:** Python, C, Kotlin, SQL, C++ (Basic)
- CS Fundamentals:** Data Structures & Algorithms, OOP, System Design Basics, REST API Design, Recursion, Sorting, Hashing, Graph Algorithms
- Backend & Frameworks:** FastAPI, MVVM, Repository Pattern, Firebase (Auth, Realtime DB), MySQL, NumPy, Pandas
- AI / ML:** PyTorch, Machine Learning, Deep Learning (CNN, RNN, Transformer), NLP, Transfer Learning, RAG, Vector DB (FAISS)
- Developer Tools:** Git, GitHub, Android Studio, Postman, VS Code, Google Colab
- Cloud & Platforms:** Google Cloud Platform (Compute, Storage, BigQuery, Vertex AI), GitHub Actions (CI/CD basics)

CERTIFICATIONS & ACHIEVEMENTS

- Google Cloud Arcade 2025 — Completed hands-on labs across GCP (Cloud Storage, BigQuery, Vertex AI).
- HackerRank Python Certification — Demonstrated proficiency in Python data structures, algorithms, and file I/O.
- Solved 270+ DSA problems on GeeksforGeeks across arrays, linked lists, binary trees, graphs, and dynamic programming.

VOLUNTEERING

- NSS (National Service Scheme) — Participated in community development initiatives, coordinating logistics and communication for a team of 20+ volunteers.